

A Composite Stress Index Framework for Quantifying Biological Stress: Theory, Ontological Structure, and Cross-Domain Instantiation

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September 1, 2026

ABSTRACT

Tumor stress phenotypes arise from coordinated activation of hypoxia, angiogenesis, glycolytic metabolism, reactive oxygen species response, and inflammatory signaling programs. These programs are commonly analyzed as independent pathway activities or combined into cohort-dependent molecular signatures, limiting interpretability, decomposability, and single-patient deployment. Here, we formalize tumor stress quantification as a deterministic mapping from high-dimensional molecular measurements into a biologically anchored stress-coordinate space. We introduce the Composite Stress Index (CSI) as a general theoretical framework for representing biological stress as a reference-standardized composite of stress-related information codes. Within oncology, this framework is specialized as the Tumor Composite Stress Index (TCSI), where the system of interest is malignant tissue and the stress axes reflect tumor-relevant adaptive programs. Beyond the theory, we built this framework as a disease-agnostic data model for stress information, structured around a three-layer stress-information ontology and aligned with FAIR (Findable, Accessible, Interoperable, Reusable) principles for cancer research data reuse.

We define the Composite Stress Coordinate (CSC) as the image of an individual transcriptomic profile under a fixed family of stress-axis enrichment maps, and TCSI as a reference-standardized scalar functional on this coordinate. Specifically, $TCSI = \tau_w \circ \Phi_R \circ \Psi \circ H(x_{raw})$, where H is preprocessing, Ψ maps a molecular profile to pathway-level stress coordinates, Φ_R standardizes these coordinates relative to a frozen reference distribution R , and τ_w projects the standardized coordinate onto a fixed nonnegative burden direction.

We establish six formal properties: axis-wise biological interpretability, exact decomposability into per-axis contributions, single-sample computability, invariance after reference fixation, monotonicity in stress coordinates, and extensibility to multiscale molecular and spatial mappings. Under standardized compound-symmetric pathway covariance, the equal-weight direction is collinear with the dominant principal-component direction; under an independent residual-noise model, the equal-weight projection reduces idiosyncratic pathway noise variance

by a factor of $1/K$. We extend the operator to spatial transcriptomics by defining location-wise stress fields and stress-boundary gradient operators. Simulation studies across six perturbation regimes characterize operator behavior under coordinated stress activation, axis-specific perturbation, reference drift, noise inflation, missing axes, and spatial heterogeneity. TCSI provides a mathematically explicit, biologically interpretable, and outcome-independent coordinate for tumor stress quantification that is deployable from a single molecular profile using a fixed reference standard.

Keywords: Tumor Composite Stress Index; Composite Stress Coordinate; reference standardization; single-sample scoring; pathway enrichment; compound symmetry; principal component analysis; spatial transcriptomics; systems oncology; FAIR data principles; stress-information ontology

1. Introduction

1.1 Clinical and Biological Motivation

The idea for StressCode emerged from a limitation encountered during preliminary pilot work. I initially attempted to integrate environmental, molecular, imaging, and outcome data to study biologically meaningful patterns across domains. However, the available data lacked sufficient overlap across these categories. Key variables were missing, inconsistently annotated, or siloed in separate datasets, resulting in a small analyzable cohort and limited statistical significance. While the pilot study was informative, it highlighted a broader infrastructure problem: publicly shared cancer research outputs are often accessible but not readily reusable for integrated biological questions. StressCode was developed to address this problem by providing a disease-agnostic stress-information framework.

Tumor microenvironmental stress arises from the coordinated activation of biological programs that enable cancer cells to survive under oxygen limitation, nutrient deprivation, immune pressure, oxidative injury, and metabolic demand. Key programs include hypoxia signaling, glycolytic metabolism, angiogenic remodeling, reactive oxygen species (ROS) response, and TNF- α /NF- κ B inflammatory signaling. These programs are biologically interdependent rather than isolated modules, motivating representation of tumor stress as a coordinated multidimensional biological state. Prior pan-cancer transcriptomic analyses support co-activation of these programs as part of a broader stress-adaptation landscape implicated in tumor aggressiveness, therapeutic resistance, and adverse clinical behavior (Hanahan, 2022; Sánchez-Vega et al., 2018).

Notably, the five canonical stress axes of the TCSI framework are not only markers of oncogenic adaptation but are also well-characterized mediators of the cellular radiation response. Hypoxia is the dominant determinant of intrinsic radiation sensitivity: hypoxic cells require approximately 2.5–3 times higher absorbed radiation doses to achieve equivalent cytotoxic effect (oxygen

enhancement ratio; Hall & Giaccia, 2019). ROS, particularly the hydroxyl radical generated by the radiolysis of water, constitute the primary mechanism of radiation-induced DNA double-strand break formation (Ward, 1988). NF- κ B inflammatory signaling is a canonical post-irradiation transcriptional response pathway (Baskar et al., 2012). Warburg-shifted glycolytic metabolism is mechanistically linked to radioresistance through ATP-dependent DNA damage repair (Meijer et al., 2012). Angiogenic remodeling under radiation is driven by vascular injury and VEGF-mediated endothelial survival signaling (Garcia-Barros et al., 2003). This biological convergence between oncogenic stress axes and radiation response programs motivates a future Radiation Stress Index (RSI) as an extension of the CSI framework to health physics and radiation science contexts.

1.2 The Problem of Cohort-Relative Scoring

A further limitation concerns portability. Gene set variation analysis (GSVA; Hänzelmann et al., 2013) uses cohort-level information from the input expression matrix when estimating pathway activity, so a sample's score may depend on the cohort in which it is embedded. Adding or removing samples can alter pathway scores for otherwise identical molecular profiles. Single-sample gene set enrichment analysis (ssGSEA; Barbie et al., 2009) partially addresses cohort dependence by computing enrichment from within-sample gene ranks. However, single-sample computability alone does not guarantee cross-study comparability: rank-based scores remain sensitive to preprocessing, gene coverage, platform effects, and calibration. A tumor stress coordinate intended for portable single-sample use therefore requires a locked gene universe, pre-specified biological axes, fixed scoring functions, and a frozen reference-standardization rule (Toro-Domínguez et al., 2025).

1.3 Mathematical Desiderata

We argue that a mathematically rigorous tumor stress coordinate should satisfy five formal desiderata: (1) axis-wise biological interpretability—each coordinate dimension is indexed by a pre-specified, biologically curated gene set; (2) exact decomposability—the scalar summary is recoverable as the sum of per-axis contributions attributable to named biological programs; (3) single-sample computability—once the scoring specification is locked, the coordinate assigned to an individual sample is independent of target-cohort composition, size, and ordering; (4) fixed-reference comparability—scores from different cohorts are interpretable on the same calibrated scale once preprocessing, gene universe, scoring functions, and reference parameters are fixed; and (5) outcome independence in construction—pathways, weights, and calibration parameters are not selected or optimized using downstream clinical outcomes. Standard pathway-scoring methods, latent factor models, and outcome-trained signatures generally do not jointly guarantee all five without additional specification.

The five desiderata apply without modification to any biological domain in which coordinated stress-response axes can be specified a priori from established biology—not only oncology.

Existing tools for quantifying cellular radiation response include outcome-trained genomic radiosensitivity signatures (Eschrich et al., 2009) and small-panel radioresponse indices, but these approaches are typically cohort-dependent, outcome-fitted, or lack formal decomposability and single-sample guarantees. No existing radiation biology composite index simultaneously satisfies all five desiderata. This gap motivates a future Radiation Stress Index (RSI) as a domain-specific instantiation of the general CSI framework within health physics and radiation science.

1.4 Contribution of This Paper

We formalize composite tumor stress quantification as a mapping problem. Let X denote a molecular measurement space and $\Pi = \{P_1, \dots, P_K\}$ a fixed family of biologically curated stress pathways. For each stress axis k , an admissible scorer S_k maps a molecular profile to a pathway-level activity score. The resulting Composite Stress Coordinate is $z_i = \Psi(x_i) = (S_1(x_i), \dots, S_K(x_i))^T$. A frozen reference-standardization map Φ_R then maps the raw coordinate to a calibrated reference coordinate z_i^* . The Tumor Composite Stress Index is the scalar functional $TCSI(x_i; F) = w^T z_i^*$, with canonical equal weights $w = (1/K) \cdot 1_K$ unless a different pre-specified weighting scheme is defined in a versioned specification.

This paper develops a formal operator by defining five mathematical objects—the molecular measurement space, the stress pathway family, the admissible stress-axis scorers, the frozen reference-standardization map, and the scalar functional—whose composition defines an operator under a locked, versioned specification F corresponding to TCSI-v1. Once F is fixed, the operator maps an individual molecular profile to a reference-standardized stress coordinate and scalar index without using target-cohort composition or any downstream clinical outcome. We establish axis-wise interpretability, exact decomposability, single-sample computability, fixed-reference comparability, monotonicity under coordinate-wise increases, and extensibility to multiscale mappings. We further establish spectral equivalence between the equal-weight TCSI direction and the first principal-component direction under standardized compound-symmetric covariance, derive an analytical noise-attenuation result under an independent residual-noise model, and extend the scalar operator to spatial transcriptomics. Section 1.5 develops the ontological structure of the CSI framework; Section 1.6 formalizes the data model perspective and its alignment with FAIR principles.

1.5 A Disease-Agnostic Ontology of Stress Information

The Composite Stress Index framework is not merely a methodology for stress biology. It is, more fundamentally, a formal theory of how stress information is organized, classified, related, and interpreted across biological scales. In the vocabulary of formal knowledge representation, an ontology is a structured specification of the entities within a domain, the categories to which those entities belong, and the semantic relationships that connect them (Gruber, 1993; Smith et al., 2007). Applied to the CSI framework, the relevant domain is biological stress response, and

the entities of interest span three organizational levels: environmental stress inputs that perturb biological systems; molecular stress-response processes by which biological systems detect, transduce, and adapt to those inputs; and observable phenotypic outputs—including imaging phenotypes and clinical outcomes—through which the state of the stressed system is made measurable.

1.5.1 The Three-Layer Architecture

Layer 1 — Environmental Stress Code (Input Domain). The first ontological layer defines the external and microenvironmental stress conditions that constitute the inputs to the biological stress-response system. In the GBM tumor microenvironment, the dominant stressors are hypoxia, vascular dysfunction, and immune pressure; in radiation biology, the dominant stressor is ionizing radiation and its downstream oxidative consequences.

Layer 2 — Molecular Stress-Response Code (Process Domain). The second ontological layer defines the molecular biological pathways through which cells and tissues detect and respond to stress inputs. In TCSI-v1, the five canonical stress axes—hypoxia signaling, glycolytic metabolic reprogramming, angiogenic remodeling, ROS response, and TNF- α /NF- κ B inflammatory signaling—constitute the five process entities. Each process entity is operationalized by a pre-specified, biologically curated gene set.

Layer 3 — Phenotypic Output Code (Observable Domain). The third ontological layer defines the observable phenotypic outputs through which stress states are expressed as measurable biological or clinical entities. These include imaging phenotypes such as contrast enhancement, perfusion abnormality, diffusion restriction; histological features; and clinical outcomes such as treatment response and radioresistance.

The three-layer structure can be represented as a directed information-flow schema: Environmental Stress Inputs → Molecular Stress-Response Pathways → Observable Phenotypic Outputs → Clinical Outcomes.

1.5.2 Ontological Roles of CSI, TCSI, and RSI

Instantiation	Ontological Role	Layer 1: Dominant Stressors	Layer 2: Molecular Process Axes	Layer 3: Primary Observables
CSI	General stress-information structure; disease-agnostic parent ontology	Any biologically specified stressor	Any a priori curated stress-response gene set family	Any validated molecular, imaging, or clinical phenotype

TCSI	Tumor-specific stress-information structure; oncology-domain instantiation	Hypoxia, vascular dysfunction, nutrient limitation, immune pressure	Hypoxia, Angiogenesis, Glycolysis, ROS response, TNF- α /NF- κ B inflammation	Transcriptomic stress coordinates; spatial stress fields; radiogenomic imaging alignment
RSI	Radiation-specific stress-information structure; health physics and radiation biology instantiation	Ionizing radiation; radiation-induced oxidative stress	Radiation-induced HIF/hypoxia, ROS/DNA damage response, post-irradiation NF- κ B, Warburg glycolysis, radiation angiogenic remodeling	R2*, DCE-MRI Ktrans, ADC, clonogenic survival, tumor control probability, radiation injury endpoints

Table 3. Ontological characterization of CSI, TCSI, and RSI across the three layers of the stress information ontology.

1.5.3 Disease-Agnostic Scope

The ontological structure of the CSI framework is explicitly disease-agnostic. GBM serves as the primary test case because it exhibits strong co-activation of all five canonical TCSI stress axes. However, the three-layer ontological architecture and the CSI mathematical operator do not depend on GBM-specific biology. The same Layer 1 $\rightarrow$ Layer 2 $\rightarrow$ Layer 3 information-flow schema applies to any biological context in which coordinated stress-response axes can be specified a priori from established domain knowledge.

1.6 CSI/TCSI/RSI as a FAIR-Aligned Data Model

The Composite Stress Index framework is simultaneously an operational data model for representing, organizing, and linking cancer research outputs across studies, institutions, and disease contexts. By specifying a fixed set of core entities, typed relationship terms, and minimum metadata requirements, CSI/TCSI/RSI provides a disease-agnostic stress-information data model—a structured vocabulary and schema capable of functioning as a cross-domain semantic bridge between genomic, imaging, clinical, and mechanistic data.

Entity Class	Definition
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Stress Source	Any exogenous or endogenous agent or condition that imposes a perturbation sufficient to elicit a coordinated stress response
Stress Signal	A measurable molecular, cellular, or physiological indicator that encodes information about the magnitude, type, or duration of stress
Stress-Response Pathway	A canonical biological cascade or regulatory network activated by stress signals
Tumor Stress Phenotype	The composite phenotypic state emerging from integration of multiple co-active stress-response pathways
Imaging Phenotype	A quantitative descriptor from diagnostic imaging serving as a non-invasive proxy for underlying biological stress
Treatment Exposure	A therapeutic intervention characterized by modality, dose, timing, and fractionation
Outcome	A measurable clinical or biological endpoint reflecting the net consequence of treatment and stress phenotype interaction
Research Output	Any data artifact, analytical result, or knowledge product generated by a study involving the above entities

Table 4. Core Entity Classes of the CSI/TCSI/RSI stress-information data model (compressed).

The data model is explicitly designed to support the FAIR data principles (Wilkinson et al., 2016). Findability is promoted by assigning persistent identifiers to each entity class. Accessibility is supported by open schema definitions compatible with standard knowledge graph technologies. Interoperability is achieved through a typed relationship vocabulary providing a controlled predicate set for asserting cross-study and cross-domain semantic links. Reusability is enabled by minimum metadata specifications ensuring that research outputs retain sufficient provenance, context, and biological annotation.

The framework interfaces with existing biomedical ontologies: disease terms resolve against the National Cancer Institute Thesaurus (NCIt); molecular pathway entities map to Gene Ontology (GO); phenotypic descriptors align with Human Phenotype Ontology (HPO); imaging descriptors are compatible with RadLex. Full tables and detailed relationship predicates are available in the full manuscript.

2. Mathematical Framework

Composite Stress Index (CSI). The CSI is a general reference-standardized framework for quantifying coordinated biological stress across tissue and disease systems. Given a locked deployment specification F , the CSI operator maps a compatible molecular profile x_{raw} to a reference-standardized scalar coordinate of coordinated stress-pathway activity. The framework requires a locked harmonized gene universe, a pre-specified family of biologically curated stress axes, a deterministic single-sample pathway-scoring operator, a frozen reference cohort from which centering and scaling parameters are derived, and a pre-specified projection vector. The framework is not restricted to cancer.

Tumor Composite Stress Index (TCSI). TCSI is the tumor-specific implementation of CSI, in which the biological system of interest is malignant tissue and the stress axes reflect tumor-relevant adaptive programs including hypoxia, glycolytic metabolism, angiogenesis, ROS response, and inflammatory signaling. TCSI is the primary object of analysis in this paper; all theoretical properties and simulation studies are stated for TCSI as an oncology-specific CSI instantiation. The mathematical structure applies to any CSI implementation under the same assumptions.

Radiation Stress Index (RSI). The RSI is the radiation-biology-specific instantiation of the general CSI framework, in which the biological system of interest is radiation-stressed tissue and the stress axes reflect the canonical molecular mediators of the cellular radiation response. The locked RSI operator is defined by $RSI(x_i; F_{\text{RSI}}) = w_{\text{RSI}}^T z_{\text{RSI},i}^* = (1/K_{\text{RSI}}) \sum (z_{\text{RSI},ik} - \mu^{\text{R}}_{\text{RSI},k}) / \sigma^{\text{R}}_{\text{RSI},k}$, where $z_{\text{RSI},ik}$ is the raw pathway score for radiation stress axis k and the standardization parameters are derived from the frozen radiation reference cohort R_{RSI} . All six theoretical properties established for TCSI hold for RSI under the same assumptions, by identical proofs with F_{RSI} substituted for F . Candidate radiation stress axes include: radiation-induced hypoxia and HIF-1 α signaling (oxygen enhancement ratio mechanism); radiation-induced ROS response and DNA damage repair (hydroxyl-radical-driven double-strand breaks); post-irradiation NF- κ B inflammatory signaling; Warburg-shifted glycolytic metabolism linked to ATP-dependent repair; and radiation-induced angiogenic remodeling. These five axes correspond directly to the five canonical TCSI domains, establishing biological and structural continuity between oncologic and radiobiological CSI instantiations. Formal RSI validation is reserved for future work.

2.1 Notation and Core Objects

Let G denote the locked harmonized gene universe ($|G| = p$ genes), with each gene identified under the HUGO Gene Nomenclature Committee standard. Let x_{raw} denote a raw

transcriptomic input and $x = H(x_{\text{raw}}) \in \mathbb{R}^p$ the harmonized expression vector produced by a fixed preprocessing and harmonization operator H . The map H locks gene-identifier mapping, duplicate-symbol handling, retention of genes in G , expression transformation (e.g., $\log_2(1 + \cdot)$), missing-gene rules, assay-compatibility criteria, and pathway-scoring input format. A frozen reference without a locked H does not guarantee cross-cohort portability.

Definition 2.1 (Locked pathway family). The canonical pathway family is $\Pi = \{P_1, \dots, P_K\}$, with $K = 5$ in TCSI-v1. Each pathway is operationalized by a curated gene set drawn from the MSigDB Hallmark collection (Liberzon et al., 2015), locked at a named release version. Pathways are selected a priori from biological stress-adaptation mechanisms and not by survival association or any downstream outcome. The parameter K is fixed by ontological choice; any alteration defines a new version.

Definition 2.2 (Admissible scorer). A map S_k is an admissible raw-input scorer for stress axis k if it is (i) deterministic and single-profile computable—its output depends only on x_{raw} , H , the locked gene set for P_k , and locked scorer parameters; (ii) monotone-consistent—if expression of genes in P_k increases uniformly in rank order, S_k does not decrease; and (iii) preprocessing-compatible with H . The canonical implementation is ssGSEA with locked parameters (rank exponent and rank normalization enabled), but the framework admits any scorer satisfying these criteria.

Definition 2.3 (Composite Stress Coordinate). For sample i with harmonized expression vector x_i , the Composite Stress Coordinate is $z_i = \Psi(x_i) = (S_1(x_i), \dots, S_K(x_i))^T \in \mathbb{R}^K$. Each coordinate is tied to a pre-specified stress axis and operationalized through a fixed gene set, distinguishing CSC from learned latent representations such as PCA, where axes are data-driven linear combinations without fixed biological meaning.

2.2 Frozen Reference Standardization and Scalar Projection

Definition 2.4 (Frozen reference manifold). Let $R = \{x_1^R, \dots, x_{n_R}^R\}$ be a fixed reference cohort of n_R samples, each processed through the same H and Ψ . For each pathway k , the frozen reference mean and standard deviation are μ_k^R and σ_k^R . Define the reference mean vector μ^R and the reference diagonal scaling matrix $D^R = \text{diag}(\sigma_1^R, \dots, \sigma_K^R)$. Assuming $\sigma_k^R > 0$ for all k , the frozen-reference standardization operator is $\Phi_R(z) = (D^R)^{-1}(z - \mu^R)$, with components $z_{ik}^* = (z_{ik} - \mu_k^R)/\sigma_k^R$. Once established from R , the parameters μ^R and D^R are frozen and are not updated when TCSI is applied to a new target cohort. This is the mathematical mechanism that achieves reference-coordinate comparability.

Definition 2.5 (Locked CSI/TCSI operator). For canonical TCSI-v1, the projection vector is the equal-weight vector $w = (1/K) \cdot 1_K$. The locked operator is $\text{TCSI}(x_{\text{raw}}; F) = \tau_w \circ \Phi_R \circ \Psi \circ H(x_{\text{raw}})$. Equivalently, for sample i , $\text{TCSI}(x_i; F) = w^T z_i^* = (1/K) \sum_{k=1}^K z_{ik}^*$. This is the primary equation of the framework. The per-axis contribution is $c_{ik} = w_k \cdot$

$z^*_{\{ik\}}$, and the exact decomposability identity holds: $TCSI = \sum_k c_{\{ik\}}$. This identity is an algebraic consequence of the linear functional form: TCSI is exactly decomposable into axis-wise contributions attributable to named biological programs.

Definition 2.6 (Versioned deployment specification). The complete locked deployment specification for TCSI-v1 is $F = (G, H, \Pi, \Psi, R, \mu^R, D^R, w)$. Given F , the score is deterministic for any compatible input. Any change in any element defines a new TCSI version and must be reported as such.

Geometry. The scalar TCSI score partitions the reference-standardized stress space into level sets: for any constant c , $L_c = \{z^* : w^T z^* = c\}$. Under equal weights these are hyperplanes orthogonal to the all-ones direction. Samples on the same level set share an identical TCSI score but may differ in individual pathway coordinates.

2.3 Assumptions of the Canonical Construction

The canonical CSC/TCSI construction rests on the following explicit assumptions. (A1) K and the gene sets Π are fixed before any model evaluation or outcome analysis. (A2) Each scorer S_k is specified before analysis and belongs to the class of admissible scorers. (A3) All pathway-axis loadings on the common stress direction are nonnegative and directionally consistent; when loadings are zero or negative, recovery of the latent stress state degrades monotonically. (A4) Canonical TCSI uses equal nonnegative weights $w_k = 1/K$, defining an outcome-blind descriptive burden functional rather than a fitted prognostic model. (A5) Each pathway gene set contains sufficient observed genes; scoring remains stable under up to approximately 25% gene missingness per pathway. (A6) TCSI is an additive marginal burden index and does not capture higher-order pathway interactions. (A7) No clinical outcome labels are used to define G, H, Π, Ψ, R , or w . (A8) Samples are treated as independent bulk tumor transcriptomic profiles; spatial and longitudinal extensions relax this assumption in defined ways.

3. Theoretical Properties

We establish six formal properties of the locked TCSI operator. Together they guarantee that the coordinate system is biologically interpretable, algebraically decomposable, single-sample deployable, reference-comparable, directionally consistent, and extensible to multiscale mappings.

3.1 Single-Sample Computability

Proposition 3.1 (Single-sample computability). Let F be fixed. For any target cohort and any individual sample x_i , the score $TCSI(x_i; F)$ is independent of target-cohort composition, size, and ordering.

Proof. By Definition 2.5, $\text{TCSI}(x_i; F) = w^T \Phi_R(\Psi(H(x_i)))$. Inspect each factor: H is a fixed map depending only on x_i and the locked specification; Ψ is deterministic and single-profile computable; μ^R and D^R are frozen parameters derived from R , independent of the target cohort; and w is a pre-specified constant vector. Therefore TCSI is a function only of x_i and F . ■

3.2 Outcome Independence in Construction

Proposition 3.2 (Outcome independence). Let Y denote any downstream clinical or biological outcome variable. If F is specified without consulting Y , then $\text{TCSI}(x_i; F)$ is outcome-independent in construction.

Proof. By inspection of $F = (G, H, \Pi, \Psi, R, \mu^R, D^R, w)$, no element is defined as a function of Y . The gene universe is drawn from a curated nomenclature standard; H is a preprocessing rule; Π is an a priori biological selection; Ψ is a deterministic enrichment operator; R is selected by biological and assay-compatibility criteria; μ^R and D^R are statistics of the reference distribution; and w is fixed in advance. None encodes information about Y . ■

3.3 Frozen-Coordinate Comparability

Proposition 3.3 (Frozen-coordinate comparability). For two compatible raw inputs x_i and x_j processed under the same locked specification F , $\text{TCSI}(x_i) - \text{TCSI}(x_j) = w^T(z_i^* - z_j^*)$. Thus TCSI differences are expressed in units of reference standard deviations along the locked pathway axes, relative to the frozen calibration baseline established by R .

3.4 Monotone Transformation Stability

Proposition 3.4 (Monotone transformation stability). Suppose Ψ depends only on within-profile gene ranks. Let ϕ be a strictly monotone transformation applied uniformly to all measured genes in a harmonized expression vector. Then, in the absence of ties and missing values, $\Psi(\phi(x)) = \Psi(x)$, and consequently TCSI is invariant under ϕ .

3.5 Large-Sample Behavior of Reference Parameters

Proposition 3.5 (Large-sample reference consistency). Let R_n be an i.i.d. draw of size n from a population distribution over compatible transcriptomic profiles, with population reference parameters μ^* and D^* . Assume $\sigma_k^* > 0$ for all k . Then as $n \rightarrow \infty$, $\hat{\mu}^R \rightarrow^p \mu^*$, $\hat{D}^R \rightarrow^p D^*$, and for any fixed compatible input x , $\text{TCSI}(x; F_n) \rightarrow^p \text{TCSI}(x; F^*)$, the population-standardized score.

3.6 Coordinate-Wise Monotonicity

Proposition 3.6 (Coordinate-wise monotonicity). Let $w_k \geq 0$ for all k . If $z_i^* \geq z_j^*$ componentwise, then $\text{TCSI}(x_i) \geq \text{TCSI}(x_j)$. That is, the TCSI functional is coordinate-wise monotone in the reference-standardized stress coordinates.

Proof. $\text{TCSI}(x_i) - \text{TCSI}(x_j) = \sum_k w_k (z_{ik}^* - z_{jk}^*)$. Since $w_k \geq 0$ and $z_{ik}^* - z_{jk}^* \geq 0$ for every k , each summand is nonnegative, so the difference is nonnegative. ■

4. Spectral Analysis and Noise Attenuation

4.1 Latent Stress-State Model

Definition 4.1 (Latent stress-state model). Assume the reference-standardized pathway vector admits the decomposition $z_i^* = \lambda \theta_i + \varepsilon_i$, where θ_i is an unobserved integrated stress state, $\lambda \in \mathbb{R}^K$ is a vector of pathway loadings, and ε_i is a vector of idiosyncratic pathway residuals with $E[\varepsilon_i] = 0$. Substituting into the TCSI operator gives $\text{TCSI}_i = (w^T \lambda) \theta_i + w^T \varepsilon_i$. Hence TCSI estimates the shared stress state up to a scaling factor, provided pathway loadings are positive and reasonably aligned.

4.2 Spectral Equivalence Under Compound Symmetry

Theorem 4.1 (Spectral equivalence). Let z^* be the frozen-reference standardized pathway vector with population covariance matrix $\Sigma = \sigma^2[(1 - \rho)I + \rho \cdot 1 \cdot 1^T]$, where $\rho > 0$. Then: (i) the leading eigenvector of Σ is $v_1 = K^{-1/2} \cdot 1_K$ with leading eigenvalue $\lambda_1 = \sigma^2[1 + (K - 1)\rho]$; (ii) all remaining eigenvectors orthogonal to 1_K have common eigenvalue $\lambda_2 = \sigma^2(1 - \rho)$; (iii) since $\lambda_1 > \lambda_2$ for $\rho > 0$, the canonical TCSI score satisfies $\text{TCSI} = K^{-1/2} \cdot \text{PC1}$. Consequently, under exact compound symmetry, equal-weight TCSI is proportional to the first principal-component score.

4.3 Idiosyncratic Noise Attenuation

Theorem 4.2 (Noise attenuation). Assume the simplified single-factor model $z_i^* = 1_K \cdot \theta_i + \varepsilon_i$, where residuals are mutually independent with $\text{Var}(\varepsilon_{ik}) = \sigma^2_\varepsilon$. Then $\text{TCSI}_i = \theta_i + w^T \varepsilon_i$, and $\text{Var}(\text{TCSI} | \theta) = \sigma^2_\varepsilon / K$. For $K = 5$, independent residual variance is reduced to 20% of the individual pathway residual variance—an 80% reduction.

Corollary 4.1 (Correlated residuals). Under compound-symmetric residual covariance with variance σ^2_ε and pairwise correlation ρ_ε , $\text{Var}(\text{TCSI} | \theta) = \sigma^2_\varepsilon [1/K + ((K - 1)/K) \cdot \rho_\varepsilon]$. For $K = 5$ this is $\sigma^2_\varepsilon [0.2 + 0.8 \rho_\varepsilon]$. If $\rho_\varepsilon = 0$, residuals are independent and residual variance is reduced to σ^2_ε / K . If $\rho_\varepsilon = 1$, residuals are fully correlated and residual variance equals σ^2_ε , so averaging provides no attenuation.

5. Spatial Extension

Spatial transcriptomics technologies generate expression profiles at spatially indexed locations within tissue sections. Each location s carries a transcriptomic profile $x(s)$, enabling molecular phenotyping while preserving tissue-coordinate information.

Definition 5.1 (Location-wise stress field). Define the unsmoothed spatial TCSI field by $\text{TCSI}(s) = \text{TCSI}(x(s); F)$. Single-sample computability (Proposition 3.1) extends immediately to each spatial location because the score depends only on the local profile $x(s)$ and the fixed elements of F .

Definition 5.2 (Kernel-smoothed field). For observed spatial locations $\{s_j\}$, define the kernel-smoothed TCSI field $\text{TCSI}_h(s) = \sum_j K_h(s, s_j) \text{TCSI}(s_j) / \sum_j K_h(s, s_j)$, using the Gaussian spatial kernel $K_h(s, s_j) = \exp(-\|s - s_j\|^2 / 2h^2)$, where h is the spatial bandwidth. Smaller h preserves local variation but may retain spot-level noise; larger h emphasizes broader tissue-scale stress patterns at the cost of spatial resolution. The bandwidth must be reported explicitly.

Definition 5.3 (Stress-boundary gradient). Define the spatial gradient $\nabla \text{TCSI}_h(s)$ of the smoothed field and the stress-boundary magnitude $\|\nabla \text{TCSI}_h(s)\|$. Large values identify locations where the molecular stress coordinate changes rapidly across tissue space.

6. Simulation Studies

Six pre-specified simulation studies evaluate the mathematical behavior, robustness, and implementation sensitivity of the CSC/TCSI framework. Unless otherwise stated, all simulations use $K = 5$ pathway axes, $n = 1,000$ samples, and random seed 42.

6.1 Simulation 1: Compound Symmetry and PC1 Equivalence

Objective: Evaluate the prediction that under exact compound symmetry, canonical TCSI is perfectly correlated with the first principal-component score. Result: Estimated correlations $r(\text{TCSI}, \text{PC1})$ for $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ were 0.996, 0.9999, 1.000, 1.000, 1.000 respectively, confirming Theorem 4.1.

6.2 Simulation 2: Unequal Loadings

Objective: Evaluate how well TCSI recovers the latent stress state θ when pathway loadings are unequal or partially misaligned. Result: Correlations $r(\text{TCSI}, \theta)$ for loading vectors (a) (1,1,1,1,1), (b) (1,1,1,0.5,0.5), (c) (1,1,0.5,0.2,0.2), (d) (1,0.8,0.4,0,-0.2) were 0.977, 0.964, 0.932, and 0.871 respectively. Performance degrades monotonically as loadings become heterogeneous, weak, zero, or negative.

6.3 Simulation 3: Independent vs. Correlated Residuals

Objective: Verify the variance formula in Corollary 4.1. Result: Relative errors between empirical and theoretical $\text{Var}(\text{TCSI} - \theta)$ for $\rho_\varepsilon \in \{0, 0.25, 0.50, 0.75\}$ were 7.82%, 3.88%, 0.24%, and 4.18% respectively—all within finite-sample sampling variability.

6.4 Simulation 4: Missing Pathway Genes

Objective: Assess sensitivity to incomplete pathway gene coverage. Result: Pearson $r(\text{TCSI_missing}, \text{TCSI_complete})$ remained > 0.95 for gene-removal fractions $q \leq 0.25$, with meaningful degradation at $q = 0.50$. This justifies a 75% gene-coverage threshold.

6.5 Simulation 5: Cross-Platform Perturbations

Objective: Evaluate stability under transformations resembling cross-platform expression differences. Result: Log transform and quantile normalization yielded $r \approx 1.000$. Gaussian noise $\sigma = 0.5 \rightarrow r > 0.99$; $\sigma = 1.0 \rightarrow r > 0.97$. Dropout $p = 0.10 \rightarrow r > 0.99$; $p = 0.25 \rightarrow r > 0.95$.

6.6 Simulation 6: Reference-Cohort Stability

Objective: Quantify sensitivity to finite-sample variability in reference cohort R. Result: Two hundred bootstrap reference cohorts ($n_R = 200$) applied to a fixed test set yielded mean Spearman rank correlation $\rho_S = 0.9996$ (SD = 0.00023). The frozen-reference construction is robust to finite-sample perturbation.

6.7 Summary

Collectively, the six studies establish: (i) spectral validity—TCSI is equivalent to PC1 under compound symmetry; (ii) a robustness boundary— $r(\text{TCSI}, \theta) \geq 0.871$ across all loading scenarios; (iii) analytical tractability—Corollary 4.1 confirmed; (iv) gene-coverage robustness—stable under up to 25% missingness; (v) platform portability—monotonic preprocessing preserves stability; and (vi) reference stability— $n_R = 200$ yields $\rho_S = 0.9996$.

Simulation	Target Property	Primary Metric	Key Result	Expectation Met
Sim 1	PC1 equivalence	$r(\text{TCSI}, \text{PC1})$	$r \rightarrow 1.000$ for $\rho \geq 0.3$	Yes
Sim 2	Unequal loadings	$r(\text{TCSI}, \theta)$	$r \geq 0.871$	Yes
Sim 3	Noise attenuation	$\text{Var}(\text{TCSI} - \theta)$	Max error $< 8\%$	Yes
Sim 4	Missing genes	r with complete	$r > 0.95$ for $q \leq 0.25$	Yes
Sim 5	Cross-platform	$r(\text{perturbed}, \text{baseline})$	Stable under monotonic	Yes
Sim 6	Reference stability	Bootstrap ρ_S	$\rho_S = 0.9996$	Yes

Table 1. Summary of six pre-specified simulation studies evaluating the CSC/TCSI framework.

7. Discussion

7.1 Summary of Contributions

This paper defines TCSI as a formal coordinate operator satisfying five mathematical desiderata: fixed axis-wise interpretability, reference-standardized single-sample computation, exact decomposition into pathway-axis contributions, monotonicity in stress coordinates, and extensibility to spatial or multiscale tissue representations. The central result is Proposition 3.1 (single-sample computability): given one compatible molecular profile and a locked specification F , the score can be computed without retraining, cohort-level recomputation, outcome labels, or refitting of latent variables. Theorem 4.1 provides spectral justification for the equal-weight canonical projection under compound-symmetric covariance. Theorem 4.2 and Corollary 4.1 characterize noise-attenuation properties and identify the residual-correlation regime. The spatial extension formalizes TCSI as a scalar field over tissue coordinates with kernel-smoothed estimation and a stress-boundary gradient operator.

Method	Coordinate meaning	Single-sample	Decomposable	Fixed biological	Reference-invariant
PCA	Learned data-driven	Only with frozen	Partly	No	Only if frozen
Latent factor	Learned latent	Model-dependent	Not generally	No/weak	Only if frozen
Topic model	Learned topics	Model-dependent	Not generally	Usually no	Only if frozen
UMAP	Learned nonlinear	No/weakly	No	No	No
Pathway-score	Fixed biological	Yes	No scalar	Yes	Depends
CSC/TCSI	Fixed biological	Yes	Yes	Yes	Yes

Table 2. Comparison of CSC/TCSI with existing coordinate representations across five mathematical properties.

7.2 Implementation Requirements

Deployment of TCSI requires locked implementation components. The frozen reference cohort R should be constructed from a pan-cancer compatible RNA-seq resource (e.g., TCGA pan-cancer) using only a priori inclusion criteria. A reference cohort of $n_R \geq 200$ samples is sufficient: bootstrap resampling demonstrated near-perfect rank stability (Spearman $\rho = 0.9996$). Cross-platform deployment requires monotonic preprocessing applied prior to pathway scoring.

Reliable pathway scoring requires that at least 75% of each canonical pathway gene set be observed. Single-sample uncertainty can be propagated analytically via Corollary 4.1.

7.3 Decision-Support Translation

TCSI is designed as an interpretable molecular coordinate that may eventually be displayed within human-in-the-loop decision-support systems, providing a structured summary of transcriptomic stress-axis activity while leaving interpretation and decision-making to qualified human users. It is not presented as an autonomous diagnostic, prognostic, or treatment-selection algorithm. The per-axis decomposition $TCSI = \sum_k c_{ik}$ enables direct visualization of each pathway's contribution. Whether such displays improve clinical reasoning is an empirical question requiring dedicated prospective evaluation.

7.4 Limitations

Several limitations bound the scope of the framework. Reference dependence: TCSI values are interpretable relative to the frozen reference distribution, not as absolute biological constants. Assay compatibility: the locked harmonization map H must be applied consistently. Pathway-set dependence: each version must lock and report exact pathway definitions. Equal-weight assumption: heterogeneous, weak, zero, or negative loadings can reduce alignment with the dominant latent stress axis (Simulation 2, scenario d: $r \approx 0.87$). Additive combination: interactive or synergistic relationships among stress programs are not captured. Spatial bandwidth: h must be reported explicitly and evaluated. No clinical utility is claimed: disease-specific validation is reserved for later manuscripts.

7.5 Relationship to Existing Methods

Elements of the TCSI framework have precedent—including single-sample pathway scoring, reference-based calibration, and tumor stress biology—but the novelty is the formal integration into a unified, closed-form operator with explicitly stated mathematical properties. CSC/TCSI combines single-sample pathway scoring, frozen reference standardization, biologically anchored stress axes, equal-weight canonical aggregation, exact per-axis decomposability, spectral justification under compound symmetry, and spatial extension. Unlike unsupervised latent factor models, TCSI has fixed axes and does not require cohort-specific refitting. Unlike supervised signatures, TCSI is not trained on outcome labels. Unlike generic pathway-score panels, TCSI specifies a reference-standardized scalar operator with mathematical properties and a versioned deployment specification.

7.6 Future Work

Paper 2 will evaluate the locked specification as a disease-specific biomarker in glioblastoma, using independent validation cohorts and potentially including Cox regression, Kaplan–Meier analysis, calibration assessment, and comparison against established prognostic variables. Paper 3 will extend the framework toward spatial and radiogenomic integration, including spatial

transcriptomic validation of TCSI fields, comparison of stress-boundary magnitude with histologic or imaging features, and alignment of molecular stress coordinates with radiologic phenotypes. Additional extensions may include alternative weighting schemes learned in strictly external training data, nonlinear or interaction-aware stress operators, platform-specific calibration transfer functions, and prospective human-factors evaluation of TCSI-based displays.

8. Conclusion

Given one compatible molecular profile and a locked TCSI specification F , the operator returns one reproducible, frozen-reference stress coordinate. This is the core thesis of the present work. TCSI is defined as a closed-form linear functional on a reference-standardized K -dimensional pathway-score vector, with pre-specified equal weights, a frozen reference distribution, and a versioned deployment specification: $\text{TCSI}(x; F) = (1/K) \sum_k (z_k - \mu_k^R) / \sigma_k^R$.

Its mathematical properties—single-sample computability, outcome independence in construction, spectral equivalence to PC1 under compound symmetry, and independent noise attenuation under $\text{Var}(\text{TCSI} \mid \theta) = \sigma_\epsilon^2 / K$ —are established by proof and characterized under generalized conditions. For $K = 5$, averaging reduces independent residual variance by 80%. More generally, under compound-correlated residuals, $\text{Var}(\text{TCSI} \mid \theta) = \sigma_\epsilon^2 [1/K + ((K - 1)/K) \cdot \rho_\epsilon]$, identifying the residual-correlation regime. The spatial extension defines TCSI as a scalar field over tissue coordinate spaces.

Across six pre-specified simulation studies, the framework demonstrated spectral equivalence to PC1, monotonically bounded degradation under loading misalignment, empirical confirmation of the closed-form variance decomposition, stability under up to 25% gene missingness, robustness to monotonic cross-platform transformations, and negligible reference-cohort sensitivity with $n_R \geq 200$. Collectively, these results establish that TCSI is theoretically grounded, analytically tractable, and robust to realistic implementation variability, supporting its deployment as a frozen, single-sample stress index without outcome-dependent calibration. The Tumor Composite Stress Index is one implementation of the broader CSI framework; the same mathematical structure applies to any tissue or disease system for which biologically motivated stress axes can be specified a priori. Together, these results establish TCSI as a mathematically rigorous, reproducible, reference-standardized molecular coordinate ready for downstream disease-specific validation and spatial-radiogenomic integration.

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